GHANA COMMUNICATION TECHNOLOGY UNIVERSITY (GCTU)

COURSE OUTLINE: CSSD 111 – INTRODUCTION TO COMPUTER SYSTEMS

Faculty: Computing and Information Systems

Department: Computer Science

COURSE CODE:	CSSD 111	**LEVEL:**	100
SEMESTER:	First Semester 2025/2026	**CREDIT HOURS:**	3
LECTURER:	Mutala Mohammed	**MOBILE/WHATSAPP:**	0244015912
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OFFICE HOURS:	Mon & Tues 9am – 5pm		

1. COURSE DESCRIPTION

CSSD 111 is an introductory course in computer concepts designed to provide students with foundational computer literacy to function effectively in today's digital and organizational environments. The course covers both theoretical and practical aspects of computing, emphasizing hardware, software, networking, digital security, databases, and system applications.

In alignment with the official course description, the course integrates personal computing knowledge with enterprise-level system awareness. It consists of two balanced components:

- Lecture Component (2 Hours Weekly): Focused on hardware, software, systems, networks, and digital ethics.
- Laboratory Component (1 Hour Weekly): Hands-on sessions using Microsoft Office Suite and system simulation tools.

2. COURSE AIM

The aim of this course is to develop digitally competent students capable of understanding, evaluating, and applying computer system concepts within both individual and organizational contexts.

3. COURSE OBJECTIVES

By the end of this course, students will be able to:

1. Explain fundamental computing concepts and digital representation.
2. Identify and analyze system hardware components and memory architecture.
3. Differentiate between system software and application software.
4. Describe operating system functionality and resource management.
5. Explain networking fundamentals and communication protocols.
6. Identify digital security threats and ethical responsibilities.
7. Demonstrate professional use of Microsoft Word, Excel, and PowerPoint.
8. Integrate theoretical and practical knowledge into an enterprise computing solution.

4. LEARNING OUTCOMES

Upon successful completion, students will be able to:

- Define and distinguish data, information, knowledge, hardware, software, system, etc.
- Describe the CPU machine cycle and memory hierarchy.
- Compare storage technologies and cloud models.
- Explain OS functionality and system utilities.
- Identify network types and simulate simple networks.
- Evaluate security risks and apply ethical computing principles.
- Develop structured reports, spreadsheets, and presentations.
- Synthesize computing knowledge into a structured enterprise proposal.

5. TEACHING & LEARNING METHODS

- Face-to-face Lectures (2 hours weekly)
- Structured Laboratory Practice (1 hour weekly)
- Demonstrations
- Guided Simulations
- Project-Based Learning

6. INTEGRATED PROJECT: SMALL BUSINESS DIGITAL STARTUP SOLUTION

[Students act as IT consultants designing a digital infrastructure for a small startup business. Each week's lab contributes to the cumulative solution including hardware selection, budgeting, networking, security, and presentation.]

Project Scenario: Students act as IT Consultants for a new local startup (e.g., Agro-Processing Factories, Organic Farms & Aggregators, Traditional "Chop Bars" & Local Food Joints, Community Bakeries (Bread & Pastry Shops), a modern pharmacy, boutique, law firm, Microfinance & Credit Unions, Private Preparatory Schools, Courier & Delivery Services, Intercity Bus Terminals, Modern Spas & Fitness Centers, Neighborhood Clinics & Diagnostic Centers, Interior Design & Furniture Showrooms, Spare Parts Dealerships (e.g., Abossey Okai), or digital marketing agency). The business needs a complete technology setup to transition from manual, paper-based processes to a professional digital enterprise.

Project Structure:

- Groups: Students will be organized into teams of 5 to 6 members.
- Objective: To design and document a full "Enterprise Solution" including hardware selection, budgeting, and a network plan.,
- Methodology: Each week, students will implement a specific milestone of this project during the lab sessions.

7. 10-WEEK SCHEDULE

Textbook:

1. **Understanding Computers Today and Tomorrow**, 16th Ed. (Morley & Parker)
2. **Technology in Action: Complete.** (Alan Evans, Kendall Martin, and Mary Anne Poatsy) (Pearson)
3. **Discovering Computers: Tools, Apps, Devices, and the Impact of Technology** (Misty E. Vermaat, Susan L. Sebok, and Jennifer T. Campbell) (Cengage Learning)

Week	2-Hour Lecture Topics	Textbook Chapter	1-Hour Lab Activities	Project Milestone
1	**Introduction to the World of Technology** • Why study/impact computer systems? • Data vs. Information • Computers, Devices, and the Web • Individual vs Organizational Computing	Morley: Ch. 1	Operating System Navigation File & Directory Management	Business Selection & Project Folder Setup
2	**Hardware: Computing Components** • CPU Architecture • Machine Cycle • Memory (RAM, ROM, Cache) • Motherboard Overview	Morley**:** Ch. 2	MS Word I: Styles, TOC, Academic Formatting	Draft Executive Summary
3	**Hardware: Storage & I/O Systems** • SSD vs HDD • Cloud Storage • Input & Output Devices • Data Preservation	Morley**:** Ch. 3 & 4	MS Word II: Tables, Mail Merge	Hardware Specification Table
4	**Software: System Software & OS** • OS Functions • Process Management • Utilities & Drivers • Cloud Coordination	Ch. 5	MS Word III: Track Changes Collaboration	Complete Documentation Draft
5	**Software: Programs & Applications** • Productivity Software • Graphics Tools • Security Tools • Software Evaluation	Ch. 6	MS Excel I: Formulas Budget Calculations	Enterprise Budget Development
6	**MID-SEMESTER EXAMINATION**	Covers: Week 1–5 lessons	Theory &/or Practical Test: Word & Hardware Identification	Project Consolidation
7	**Networks and the Internet: Networks and Communication Devices & The Internet** • LAN, WAN • Network Topologies • TCP/IP Basics • Internet Infrastructure	Ch. 7 & 8	Cisco Packet Tracer Network Simulation	Startup Network Map
8	**Networks and the Internet: Security, Privacy, Digital Security & Ethics** • Malware • Encryption • Privacy	Ch. 9	MS Excel II: Logical Functions Pivot Tables	Security Cost Analysis

	• Ethical Issues			
9	**Systems: Information Systems, Database & System Development** • Information Systems • Relational Databases • AI & Automation • ***Certifications & Careers***	Ch. 10–12	PowerPoint Professional Presentation	Investor Pitch Preparation
10	**REVISION WEEK**	Comprehensive Review Chapters	Project Finalization Presentation Rehearsal	Final Submission

8. ASSESSMENT AND GRADING

Component	Weight (%)	Description
Attendance & Participation	5%	Engagement in lectures and labs.
Lab Assessments/Quizzes	15%	Weekly MS Office tasks and hardware quizzes.
Mid-Semester Exam (week 6)	20%	Theory covered in Weeks 1–5.
Practical Project (Integrated Solution)	20%	Final integrated project using Word, Excel, and PPT.
Final Examination	40%	Comprehensive theory and system concepts.

9. GCTU GRADING SYSTEM

Score (%)	Grade	Interpretation	Grade Point
80 – 100	A	Outstanding	4.00
75 – 79	A-	Excellent	3.75
70 – 74	B+	Very Good	3.50
65 – 69	B	Good	3.25
60 – 64	B-	Fairly Good	3.00
55 – 59	C+	Above Average	2.75
50 – 54	C	Average	2.50
45 – 49	C-	Below Average	2.00
40 – 44	D	Pass	1.50
0 – 39	F	Fail	0

10. RECOMMENDED PRACTICAL TOOLS

- Microsoft (Word, Excel, PowerPoint)
- Cisco Skills for All: "Computer Hardware Basics" for hardware identification.
- Cisco Packet Tracer: Used in Week 7 for network visualization.
- Wireshark: Used in Week 7 to demonstrate data encapsulation and security.
- Microsoft Learn: Alignment with MOS (Microsoft Office Specialist) certification standards.